| Part | Description | Datasheet |
|---|---|---|
| **Purchased on Breakout Board** | | |
| PSoC6 MCU | Microcontroller | https://www.digikey.com/en/products/detail/infineon-technologies/CY8CPROTO-063-BLE/9739909?_gl=1*12pozst*_up*MQ..*_gs*MQ..&gclid=CjwKCAiAraXJBhBJEiwAjz7MZVoww4bmVaP0nj-rymsyQ5x50sekqK_qqaQN0pepBzjeXeYvosMskhoCVdEQAvD_BwE&gclsrc=aw.ds&gbraid=0AAAADrbLljulcCZlq90u9Z8aMla88ISX |
| MAX30102 | Heartrate Sensor | https://www.digikey.com/en/products/detail/sunfounder/ST0244/22116824?_gl=1*nddc8g*_up*MQ..&gclid=CjwKCAiAraXJBhBJEiwAjz7MZVoww4bmVaP0nj-rymsyQ5x50sekqK_qqaQN0pepBzjeXeYvosMskhoCVdEQAvD_BwE&gclsrc=aw.ds&gbraid=0AAAADrbLljulcCZlq90u9Z8aMla88ISX |
| BMI270 | Acelerometer / GyroScope | https://www.digikey.com/en/products/detail/sparkfun-electronics/22397/21703804?_gl=1*19u0pml*_up*MQ..&gclid=CjwKCAiAraXJBhBJEiwAjz7MZVoww4bmVaP0nj-rymsyQ5x50sekqK_qqaQN0pepBzjeXeYvosMskhoCVdEQAvD_BwE&gclsrc=aw.ds&gbraid=0AAAADrbLljulcCZlq90u9Z8aMla88ISX |
| **Not Purchased** | | |
| XC620P182MR-G | Voltage Regulator (3.3 to 1.8) | https://product.torexsemi.com/system/files/series/xc6206.pdf |
| TYPE-C-31-M-12 | USB-C | https://lcsc.com/product-detail/USB-Connectors_Korean-Hroparts-Elec-TYPE-C-31-M-12_C165948.html |
| AMS117-3.3 / NCP167AMX330TBG from ON Semi | Voltage Regulator (5V to 3.3) | AMS1117-3.3 / https://www.onsemi.com/pdf/datasheet/ncp167-d.pdf |
| MCP73831 | Battery Charger | https://ww1.microchip.com/downloads/en/DeviceDoc/MCP73831-Family-Data-Sheet-DS20001984H.pdf |
| DW01 | Battery Protector | https://cdn.sparkfun.com/assets/learn_tutorials/2/5/1/DW01-P_DataSheet_V10.pdf |
| PTS810SJG250SMTRLFS | Buttons | https://www.onlinecomponents.com/en/datasheet/pts810sjg250smtrlfs-44763522/ |
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| Issues: | Main Points | |
| Battery need a ADC to track charge? | Clock: good, esp 32 has internal RTC | |
| LDO for both heartrate and aceelerometer? need 5.5v -> 3.3v | Power: handled by battery and usb-c | |
| | Rst : need to look into buttons and prog | |